

WaRF: Weighted n -Agent Reliability Framework

Technical Reference

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1 Overview

WaRF is a multi-agent code review framework that combines Bayesian weight calibration with sequential hypothesis testing to produce a single APPROVE/REJECT decision with a calibrated

confidence level.

A pool of n specialised LLM agents votes independently on the submitted code. Each vote updates a cumulative evidence score Λ via the Sequential Probability Ratio Test (SPRT). The framework stops as soon as Λ crosses a decision boundary, or after all n agents have voted. Agent reliability weights $\omega \in [0, 1]$ are per-agent and per-domain; they are learned online from oracle-labelled reviews via a Bayesian update rule.

2 Agent Pool

2.1 Personas

The default pool contains five LLM agents with fixed system prompts:

Agent	Evaluation focus	Domain strengths
AUDITOR	Compliance audit, neutral	Security, Logic
BOUNDARY	Edge cases, missing guards	Logic
ADVOCATE	Author intent, false-positive resistance	General
SKEPTIC	Adversarial reasoning, threat surface	Security
TEST-FOCUSED	Coverage, missing tests	Logic

Each agent receives: (1) a system prompt defining its persona, (2) the code under review, and (3) an instruction to respond with exactly one word (APPROVE or REJECT) on the first line, followed by free-text reasoning. Agent responses are parsed for the vote; reasoning is logged and shown in the dashboard but is not consumed by the stopping rule.

An optional sixth voter, **Bandit** (a deterministic SAST analyser), is available in the SECURITY domain via the `--sast` flag. It is assigned a fixed weight $\omega = 0.75$ and does not participate in the Bayesian calibration cycle.

2.2 Domain Classification

Agents are indexed to three review domains:

Security Authentication, injection, cryptography, input validation, serialisation.

Logic Correctness, edge cases, type safety, control flow, missing guards.

Performance Algorithmic complexity, blocking calls, memory allocation, N+1 patterns.

The domain is specified by the caller via `--domain`. Agent weights are domain-specific: an agent’s LOGIC weight is entirely independent of its SECURITY weight, capturing that BOUNDARY is more reliable on logic defects than on security ones, and SKEPTIC is the reverse.

3 Sequential Probability Ratio Test

3.1 Cumulative Evidence Score

After each agent vote, WaRF updates the log-likelihood ratio:

$$\Lambda_k = \Lambda_{k-1} + s_k \cdot \omega_k \cdot \ell \quad (1)$$

where:

- k is the vote index (agents are sampled in a fixed order)

- $s_k \in \{+1, -1\}$ is the vote sign (APPROVE = +1, REJECT = -1)
- ω_k is the calibrated weight of agent k in the active domain
- $\ell \approx 1.253$ is the base log-likelihood ratio constant
- $\Lambda_0 = 0$

The LLR increment is asymmetric in the implementation. With $P_1 = 0.80$ and $P_0 = 0.30$ encoding prior beliefs about agent reliability:

$$\ell_{\text{APPROVE}} = \ln \frac{P_1}{P_0} = \ln \frac{0.80}{0.30} \approx +0.981 \quad \ell_{\text{REJECT}} = \ln \frac{1 - P_1}{1 - P_0} = \ln \frac{0.20}{0.70} \approx -1.253$$

Because $|\ell_{\text{REJECT}}| > |\ell_{\text{APPROVE}}|$, unanimous REJECT evidence accumulates faster than unanimous APPROVE evidence. In the remainder of this document, $\ell \approx 1.253$ denotes the REJECT magnitude and is used to reason about boundary reachability in the REJECT direction; the APPROVE direction requires a larger pool or higher ω to compensate.

3.2 Decision Boundaries

WaRF uses asymmetric SPRT boundaries derived from target error rates:

$$A = \ln \left(\frac{1 - \beta}{\alpha} \right) = \ln \left(\frac{0.90}{0.05} \right) \approx +2.890 \quad (\text{APPROVE boundary}) \quad (2)$$

$$B = \ln \left(\frac{\beta}{1 - \alpha} \right) = \ln \left(\frac{0.10}{0.95} \right) \approx -2.251 \quad (\text{REJECT boundary}) \quad (3)$$

with $\alpha = 0.05$ (false-alarm rate) and $\beta = 0.10$ (miss rate). The asymmetry $|A| > |B|$ is deliberate: requiring stronger evidence for APPROVE than for REJECT. A false rejection triggers human review, which is recoverable. A false approval on a defective change is not.

3.3 Stopping Rule

After vote k :

- If $\Lambda_k \geq A$: **stop**, decision = APPROVE
- If $\Lambda_k \leq B$: **stop**, decision = REJECT
- Otherwise: invite the next agent (or proceed to synthesis)

If the pool is exhausted without crossing a boundary, WaRF uses a synthesis threshold $\theta = 0.2$:

$$\text{decision} = \begin{cases} \text{APPROVE} & \text{if } \Lambda_n > \theta \\ \text{REJECT} & \text{if } \Lambda_n < -\theta \\ \text{INCONCLUSIVE} & \text{if } |\Lambda_n| \leq \theta \end{cases} \quad (4)$$

3.4 Maximum Reachable Evidence and Pool Size

With a pool of n agents voting unanimously in one direction:

$$|\Lambda|_{\max} = \ell \cdot \sum_{k=1}^n \omega_k \quad (5)$$

SPRT early stopping is only possible if $|\Lambda|_{\max}$ exceeds the relevant boundary. Table 1 shows reachability for representative configurations.

Configuration	n	$\sum \omega_i$	$ \Lambda _{\max}^{\text{rej}}$	REJECT?	APPROVE?
Uniform $\omega = 0.5$	2	1.000	1.253	No	No
Top-2 current ($\omega \approx 0.605$)	2	1.211	1.517	No	No
Top-3 current	3	1.811	2.268	Yes	No
Top-4 current	4	2.332	2.922	Yes	Yes
Full 5-agent pool	5	2.803	3.511	Yes	Yes

Table 1: SPRT boundary reachability at current calibrated weights ($|\lambda_B| = 2.251$, $\lambda_A = 2.890$). $|\Lambda|_{\max}^{\text{rej}}$ uses $\ell_{\text{REJECT}} = 1.253$; the APPROVE direction (using $\ell_{\text{APPROVE}} = 0.981$) requires $n \geq 4$ at current weights.

At current calibrated weights, $n = 3$ is the minimum pool that can reach the REJECT boundary; $n = 4$ for the APPROVE boundary. With $n = 2$, the framework always exhausts the pool and synthesizes, capped at MEDIUM confidence. These thresholds are weight-dependent: as agents accumulate more oracle labels and $\hat{\omega}$ rises toward 1, smaller pools become sufficient.

4 Bayesian Weight Calibration

4.1 Model

Each agent a maintains a per-domain Beta distribution over its reliability:

$$\omega_a^{(d)} \sim \text{Beta}(\alpha_a^{(d)}, \beta_a^{(d)}) \quad (6)$$

The prior is $\alpha_0 = \beta_0 = 1$, the uniform distribution, corresponding to $\omega = 0.5$ with maximum uncertainty (two pseudo-observations).

4.2 Update Rule

After an oracle-labelled review in domain d , for each agent a that voted:

- Vote matched oracle verdict: $\alpha_a^{(d)} += 1$
- Vote did not match oracle verdict: $\beta_a^{(d)} += 1$

The point estimate used in Equation 1 is the posterior mean:

$$\hat{\omega}_a^{(d)} = \frac{\alpha_a^{(d)}}{\alpha_a^{(d)} + \beta_a^{(d)}} \quad (7)$$

Agents that are skipped by SPRT early stopping do not receive an update — only agents that actually voted are evaluated against the oracle.

Interpretation. $\hat{\omega}$ is an estimate of *agent-domain fit*, not of the agent’s absolute quality. An agent that performs well on LOGIC code but poorly on SECURITY code will accumulate high $\hat{\omega}^{\text{LOGIC}}$ and low $\hat{\omega}^{\text{SECURITY}}$; neither value characterises the agent in isolation. Changes in $\hat{\omega}$ over time signal that the current corpus is shifting in ways that suit or penalise the agent’s evaluation heuristics — not that the agent itself has changed.

4.3 Confidence Intervals

A 95% credible interval for ω is computed via the Normal approximation to the Beta posterior:

$$\hat{\omega} \pm 1.96 \sqrt{\frac{\hat{\omega}(1-\hat{\omega})}{\alpha + \beta}} \quad (8)$$

Wide intervals indicate few oracle-labelled reviews in the domain; the weight is not yet reliable and should be interpreted with caution.

4.4 Oracle Labelling

The oracle is human judgment: after a review session, a developer marks the ground-truth verdict. WaRF compares each agent’s vote to the oracle and updates weights accordingly. The system credits the correct *outcome*, not the quality of reasoning: a correct vote for a spurious reason receives the same reward as a correct vote for the right reason.

This is a deliberate black-box abstraction. Evaluating reasoning quality would require a developer to read and judge every explanation — which is exactly the work WaRF is meant to reduce.

5 Confidence Classification

WaRF assigns one of three confidence labels to each decision:

Label	Condition	Empirical accuracy	Sessions
HIGH	SPRT boundary crossed ($ \Lambda \geq \lambda_B $ or $\geq \lambda_A$)	75%	12
MEDIUM	Pool exhausted, $ \Lambda_n > \theta$	63%	35
LOW	Synthesis, $ \Lambda_n \leq 2\theta$	27%	15

Accuracy rises monotonically from LOW (27%) to HIGH (75%) across 62 oracle-labelled sessions, confirming that the SPRT boundary marks a meaningful evidence threshold. The LOW anti-correlation (worse than random) indicates that when the pool exhausts without accumulating much signal, the synthesis decision is actively unreliable; such cases should always route to human review.

The practical utility of the HIGH label is in *selective automation*: a downstream policy can auto-approve HIGH-confidence APPROVE decisions, block on HIGH-confidence REJECT, and escalate MEDIUM or LOW to a human reviewer. This stratification is only possible because the SPRT boundary produces a qualitatively distinct evidence tier, not merely a higher vote count.

6 Debate Round

When a Round-1 decision has LOW or MEDIUM confidence, WaRF can optionally invoke a debate round. Each agent receives all other agents’ Round-1 votes and reasoning and is asked whether it maintains or revises its position.

6.1 Prompt Design

The debate prompt is not neutral. Two framings produce systematically different outcomes on the same code:

Prompt	Typical outcome	Mechanism
<i>“Do you revise your position?”</i> (no anchor)	Conformity	Minority reads majority as evidence and capitulates
<i>“Your Round-1 vote was X. Maintain or revise? Explain what others got wrong if you maintain.”</i>	Rational persuasion	Minority has a path to defend its position

The anchoring prompt is the current default in WaRF. The conformity nudge was an early design that was found to amplify the wrong majority — including with HIGH confidence after SPRT fired on the conforming votes.

6.2 When Debate Helps and When It Does Not

Debate propagates correct reasoning that agents can evaluate and find convincing. It does not cure false beliefs held by an agent whose training does not include the relevant ground truth.

Case	Outcome	Reason
<code>sessions_pre_auth_stripping.py</code>	Corrected	Correct minority (SKEPTIC) had a verifiable argument about <code>rebuild_auth</code> stripping; majority evaluated and updated
<code>hooks.py</code>	No change	Incorrect agent (SKEPTIC) held a false belief about <code>collections.abc.Callable</code> ; correct explanation did not penetrate
<code>_internal_utils.py</code>	Worsened	Plausible wrong argument (“no None guard”) captured the correct minority; SPRT fired on the wrong unanimous verdict

Key principle: persuasion $\neq$ correctness. Debate propagates the most *persuasive* argument, not the correct one. A structurally sound generic heuristic (“no None guard”) can capture the correct minority even when it is inapplicable in context. When this happens, the result is a unanimous wrong verdict with HIGH confidence — the worst failure mode, because the SPRT boundary provides false assurance. The oracle is the only in-system corrective; debate has no mechanism to distinguish a persuasive argument from a true one.

7 Empirical Results

Results are drawn from 246 sessions across PSF/requests library code and manually constructed validation files (M-00 through M-13).

7.1 Calibration

On 62 oracle-labelled sessions (reliability diagram):

Confidence	Sessions	Correct	Accuracy
HIGH	12	9	75.0%
MEDIUM	35	22	62.9%
LOW	15	4	26.7%
Overall	62	35	56.5%

The 48-percentage-point spread from LOW (27%) to HIGH (75%) validates that the SPRT boundary marks a meaningful evidence threshold. Overall accuracy of 56.5% reflects a corpus balanced between straightforward and hard cases; the confidence-stratified numbers are the informative signal. Compared to the previous corpus snapshot, the MEDIUM band has grown (22 to 35 sessions) while LOW has shrunk (28 to 15), consistent with agents accumulating enough calibration history to escape the inconclusive range on more reviews.

7.2 Token Efficiency

SPRT early stopping fires on approximately 44% of sessions (109 of 246). On those sessions the average token usage drops substantially compared to a full-pool run:

Configuration	Tokens	Confidence	Notes
5-agent, pool exhausted	~16 100	MEDIUM (max)	All 5 agents called
5-agent, SPRT at $k = 3$	~3 200	HIGH	2 agents skipped
2-agent, any outcome	~4 000	MEDIUM (max)	Boundaries unreachable

Token cost is more strongly driven by file size than pool size; a 5-agent run on a small file is cheaper than a 2-agent run on a large one. The early stopping rate of 44% has grown with corpus size: as agents accumulate oracle labels and $\hat{\omega}$ converges away from the uninformative prior, evidence accumulates faster per agent, making boundary crossings more frequent.

7.3 2-Agent Pool Comparison

The top-2 agents by current calibrated $\hat{\omega}$ (ADVOCATE and AUDITOR, both ≈ 0.605) are structurally capped at MEDIUM confidence (Section 3.4, Table 1): their combined $|\Lambda|_{\max} \approx 1.52$, well below the REJECT boundary at 2.251. This cap holds for any choice of two agents at current weight distributions.

The critical practical consequence is *selective automation*. A downstream CI policy can only stratify decisions if confidence levels are qualitatively distinct: HIGH to auto-approve or auto-block; MEDIUM and LOW to escalate. With a 2-agent pool, every decision returns MEDIUM regardless of how unambiguous the evidence is. The routing signal disappears. The 5-agent pool produces HIGH on 44% of reviews — those are the sessions that can be handled without human intervention.

Counter-argument to [4]: their finding concerns mathematical reasoning tasks, where the defect is derived from premises. Code review is different: the defect is present or absent in the artefact, and the ensemble’s value is not deliberation but accumulating the evidence mass required to cross the SPRT boundary with HIGH confidence. The 2-agent pool achieves comparable accuracy on average but never produces the HIGH-confidence tier that makes selective automation possible.

8 Known Limitations

Type 1 — Persona bias. Agents with affirmative personas (ADVOCATE, AUDITOR) evaluate whether the code does what its author intended — not whether what the author

intended is sufficient. They correctly assess that the logic is coherent and then stop. They do not ask what happens when an integer overflows, or when a pointer is null, because those scenarios lie outside the framing their persona uses to read code. BOUNDARY and TEST-FOCUSED partially correct this by design, but any pool dominated by affirmative personas inherits the bias structurally. More agents of the same persona type amplify rather than correct it.

Type 2 — Shared knowledge gap. When all agents share the same training-level blind spot, WaRF produces a unanimous wrong verdict with HIGH confidence. CVE-2024-47081 is the canonical example: all five agents approved the vulnerable `urlparse` hostname parsing line. Adding more agents with the same training coverage amplifies the wrong signal. Bandit (SAST) partially mitigates this for patterns in its ruleset.

Type 3 — False-consensus capitulation. In debate, a correct minority can be captured by a plausible wrong majority argument (see `_internal_utils.py` case). Once SPRT fires on the conforming votes, there is no in-session recovery; the oracle update is the only corrective mechanism.

Boundary reachability. At current calibrated weights, $n \geq 3$ is required to reach the REJECT boundary and $n \geq 4$ for the APPROVE boundary (Section 3.4, Table 1). With $n = 2$, the framework always exhausts the pool and synthesizes at MEDIUM. As agents accumulate more oracle labels and $\hat{\omega}$ rises, smaller pools may become sufficient; conversely, a domain with few labelled sessions keeps $\hat{\omega}$ near the prior (0.5), making boundaries harder to reach. Adding more agents increases cost proportionally whenever the boundary is not crossed before pool exhaustion.

Oracle dependency. Bayesian calibration requires oracle-labelled reviews. Without labelling, all weights remain at their prior ($\hat{\omega} = 0.5$) and the SPRT reduces to an unweighted vote counter. The quality of calibration is bounded by the labelling rate.

9 Related Work

ConSol [1] applies SPRT to a single language model at varied sampling temperature to determine when the model has converged on a consistent answer. WaRF’s extension: heterogeneous agent personas with domain-specific learned weights. Temperature variance gives noise around one perspective; different personas give genuinely different evaluation heuristics (compliance audit vs. adversarial probing). The domain-specific ω weights are only meaningful because the agents are actually different.

Asymptotically Optimal Sequential Testing [2] derives optimal stopping policies for LLM ensembles under a combined cost-plus-wait objective. The optimal policy uses at most two LLMs. The objective differs from WaRF’s: WaRF optimises calibrated accuracy at minimum agent calls without a cost term, and does not assume homogeneous agents.

Wisdom and Delusion of LLM Ensembles [3] demonstrates that ensembles do not always outperform the single best agent, particularly when agents share training data and therefore share blind spots. This is consistent with WaRF’s Type 2 failure mode and validates the distinction between *persona diversity* (which WaRF provides) and *knowledge diversity* (which requires external tools or data augmentation).

Single-Agent Outperforms Multi-Agent Under Equal Token Budget [4] finds that on mathematical reasoning tasks, adding agents under a fixed budget degrades performance. Code review differs: the defect is not derived from premises but observed in the artefact, and the ensemble’s value lies in accumulating sufficient evidence mass to cross the SPRT boundary — not in deliberation.

References

- [1] ConSol: Sequential Consistency for LLM Outputs via SPRT. arXiv:2503.17587, 2025.
- [2] Asymptotically Optimal Sequential Testing for LLM Ensembles. arXiv:2604.01086, 2026.
- [3] The Wisdom and Delusion of LLM Ensembles. arXiv:2510.21513, 2025.
- [4] Single-Agent Outperforms Multi-Agent Under Equal Token Budget. arXiv:2604.02460, 2026.